ICSC v0.1 — Specification Audit

Implicit Communication and Substrate Coupling
Audit of the spec itself: where tests can PASS while the measurement stays INVALID.
Co-authored. Status tags: OBSERVED / DERIVED / PROPOSED. `*` = thin volume (more `*` = thinner).

F. Specification defects (read first — they drive A, B, G)

```
F1  HYPOTHESIS_NEUTRALITY is unachievable as written
    which resources get instrumented, sample rate, bin width,
    alphabet, history length = hypotheses about the channel
    → acquisition always embeds hypotheses
    FIX: neutrality → DECLARED priors + DECLARED blind spots
         acquisition manifest states what channel classes are
         structurally invisible to this sensor set
    status: DERIVED

F2  detection floor is ENCODER-RELATIVE
    planted encoder calibrates sensitivity for ITS modulation
    (timing / rate / amplitude / band / symbol duration)
    a natural channel using another modulation can sit
    anywhere relative to that floor
    → tests 4, 5 pass; NOT_DETECTED still invalid
    FIX: encoder FAMILY sweep → floor is a function over
         encoding classes, not a scalar
         NOT_DETECTED scope = covered classes only
    status: DERIVED

F3  planted encoder perturbs the substrate it calibrates
    running the encoder changes source load
    → calibration conditions ≠ natural conditions
    FIX: load-matched null encoder (same resource draw,
         zero information) run alongside
    status: DERIVED

F4  "capacity" is mislabeled
    planted encoder gives ACHIEVABLE RATE under that code,
    a lower bound — not Shannon capacity
    capacity assumes known noise stats + memoryless channel;
```

substrate channels have memory and nonstationary noise
FIX: rename axis → ACHIEVABLE_RATE (lower bound, code-specific)
status: DERIVED

F5 status vocabulary mixes two axes
provenance: OBSERVED / DERIVED / ESTIMATED
outcome: UNKNOWN / UNRESOLVED / NOT_TESTED /
BELOW_DETECTION_FLOOR
a result can be DERIVED and UNRESOLVED at once
→ "no status substitutes for another" is unsatisfiable
with one field
also missing from §5: DETECTED, NOT_DETECTED (used in §4C),
the PREDICTIVE_DIRECTION_* set (§4B)
UNKNOWN vs UNRESOLVED: no defined difference
FIX: two fields per result; define every outcome state
by its exit condition (see G)
status: OBSERVED (internal contradiction in spec)

F6 retention rule conflicts with normal data hygiene
outlier rejection / artifact removal = analysis deciding
retention
FIX: retention rules frozen in manifest before run;
raw stays immutable; every exclusion logged with rule ID;
every analysis rerunnable on unfiltered raw
status: DERIVED

F7 taxonomy-as-output needs an input the instrument never
acquires
covert vs side vs emergent hinges on ORIGIN of the encoding
(designed / learned under selection / incidental)
six axes measure behavior, not origin
functional selection shows selection, not who or what
selected
→ REPORTING cannot derive covert/side from the axes alone
FIX: separate PROVENANCE input class (design docs, training
history, config lineage) — external, tagged, optional;
without it taxonomy output = UNRESOLVED by rule
status: DERIVED

F8 clock/timebase is not specified anywhere
clock skew, jitter, timestamp quantization can CREATE or
FLIP lagged TE direction
FIX: common timebase or measured skew per run;
TE reported as a function of assumed offset
status: DERIVED

F9 "isolated" + adaptive agents

```
if agents adapt, the instrument and interventions are part
of their environment → they may re-route around them
reproducibility failure may BE the phenomenon
FIX: agent state frozen vs adapting = declared condition
status: DERIVED
```

A. Corrected architecture

```
ICSC/
├── manifest/            locked + hashed BEFORE any run
│   ├── acquisition_config      sensors, fs, resolution, timebase
│   ├── blind_spot_declaration  channel classes invisible to config
│   ├── retention_rules         exclusion rules w/ IDs
│   ├── primary_analyses        pre-declared tests, lags, bins
│   ├── tolerances              reproducibility bounds
│   └── encoder_family          modulation classes for calibration
│
├── acquisition/         append-only raw, no analysis imports
│   ├── raw_store               immutable, checksummed
│   ├── timebase                skew/jitter log per run
│   └── run_epoch               config hash; config change → new epoch
│
├── calibration/         separate runs, feeds REPORTING only
│   ├── encoder_family_sweep    floor(class, n, fs, bin, noise)
│   ├── null_encoder            load-matched, zero information
│   ├── observer_load_sweep     duty-cycle dose-response
│   └── timebase_check
│
├── analysis/
│   ├── axes/                   one module per axis, own null
│   │   ├── achievable_rate
│   │   ├── natural_transfer        MI + surrogate null
│   │   ├── predictive_direction    TE + time-shift surrogate
│   │   ├── causal_structure        interventions + placebo
│   │   ├── functional_selection    load-matched dummy receiver
│   │   └── residual_dependence     conditional surrogate
│   ├── surrogates/             block / circular-shift / IAAFT
│   ├── sensitivity/            bin, lag, k sweeps → distributions
│   └── multiplicity/           forking-path ledger + correction
│
├── provenance_input/    EXTERNAL, optional, tagged (F7)
│
└── reporting/
    ├── axis_vector             never summed, never ranked
```

```
    ├── status                  {provenance} × {outcome}
    ├── taxonomy                derived; UNRESOLVED w/o provenance
    └── unresolved_ledger       residual dependence, open paths
```

Data flow is one-way: manifest → acquisition → analysis → reporting.
Calibration → reporting only. Nothing flows back into acquisition inside an epoch.

B. Acceptance-test matrix

Each test carries a `PASS_WHILE_INVALID` line: how it can pass with the measurement still broken.

```
T01  acquisition invariant to classification labels
  input   identical source/receiver run, label set A vs B
  proc    run acq twice; swap reporting labels; diff raw checksums
          + static check: acquisition/ imports nothing from
            analysis/ or reporting/
  inv     raw_A == raw_B (bitwise, modulo declared timing noise)
  fail    any raw diff traceable to label state
  out     checksum pair, import graph
  deps    manifest lock
  limit   doesn't catch HUMAN tuning between runs
  PASS_WHILE_INVALID  operator changes fs/bins after seeing
          results → caught only by T15 (epoch lock)


T02  randomized source vs correlated background
  input   source X randomized (logged seed) + background workload
          with known periodicity
  proc    est I(X;Y) for X_randomized; est I(B;Y) for background
          compare to surrogate nulls
  inv     X-attributable dependence separable from B-attributable
  fail    randomized-X dependence vanishes when B is regressed out
          but B itself was the carrier
  out     two dependence estimates + nulls
  deps    T03, T08
  limit   only background sources that are MEASURED
  PASS_WHILE_INVALID  randomization mechanism loads substrate
          (F3) → X "transfers" via its own compute cost


T03  MI null calibration under independence
  input   independent X, Y generated with REALISTIC autocorrelation
          (AR(p), 1/f, bursty), not iid only
```

proc for M sims: est MI; surrogate null (block / circular shift / IAAFT); record rejections
inv false-positive rate ≈ declared alpha across ALL autocorrelation regimes
fail FPR > alpha + tolerance in any regime
out FPR table by regime × estimator × bin
deps none
limit only regimes simulated
PASS_WHILE_INVALID validated on iid only; naive shuffle destroys autocorrelation → real data FPR inflated

T04 detection floor from planted encoder
input encoder family E = {timing, rate, amplitude, band, ...} × symbol rates × durations
proc for e in E: sweep injected rate; find min rate with detection prob ≥ p_target
inv floor(e) finite and reported per class
fail single scalar floor reported
out floor table over (class, n, fs, bin, noise, observer)
deps T05, T10, null_encoder
limit classes outside E: floor UNKNOWN
PASS_WHILE_INVALID one encoder class → floor meaningless for natural channel in another class (F2)

T05 floor responds to conditions
input fixed encoder; vary n, fs, bin, noise level
proc recompute floor per condition
inv floor monotone in the expected direction for n, noise; aliasing visible when fs < 2·f_signal
fail floor invariant to n or noise (estimator not using data)
out floor surface
deps T04
limit monotonicity is expected, not guaranteed for bins*
PASS_WHILE_INVALID floor responds correctly but encoder load perturbs substrate (F3)

T06 TE never auto-classifies causal
input any TE result
proc static check: no code path from TE status → causal_structure status
inv causal status set ONLY by intervention module
fail any write path TE → causal
out dependency graph
deps none

```
  limit   code-level only; doesn't stop a human reading TE causally
  PASS_WHILE_INVALID  report wording ("drives", "influences")
          leaks causal reading → add lexical lint on report text


T07  common-driver exposes false direction
  input   sim: D → X at lag a, D → Y at lag b (a < b), no X→Y
          + variant with clock skew s
  proc    TE(X→Y), TE(Y→X); conditional TE given D; sweep s
  inv     unconditioned TE flags spurious X→Y;
          conditioned TE removes it when D measured;
          report shows direction flips with s
  fail    spurious direction survives conditioning, or skew
          effect not reported
  out     TE(s) curves
  deps    T03, timebase
  limit   unmeasured D cannot be conditioned out
  PASS_WHILE_INVALID  sims only include drivers designer
          thought of


T08  intervention separates common cause vs mediator
  input   sims of G1: Z→X, Z→Y   G2: X→Z→Y   G3: both
  proc    randomize X; also run PLACEBO intervention
          (same resource cost, no information content)
          compare Y response: real vs placebo
  inv     G1: no Y response beyond placebo
          G2: Y responds to X content, not placebo
  fail    cannot separate G2 from "intervention load → Z → Y"
  out     identified graph set + assumptions list
  deps    T02, null_encoder
  limit   exclusion restriction assumed; SUTVA violated by design
  PASS_WHILE_INVALID  no placebo → intervention cost itself
          is the carrier


T09  graded interference keeps axes separate
  input   noise injection on Z at levels 0..k
          + energy/load-matched placebo injection
  proc    at each level: measure transfer AND task performance
          separately; monitor agent re-routing
  inv     transfer(level), performance(level) are separate
          curves; no shared estimator
  fail    performance loss used as proxy for transfer
  out     two dose-response curves + adaptation log
  deps    T04, T10
  limit   adaptive agents may move channel to another Z' (F9)
```

PASS_WHILE_INVALID noise injection itself contends for
        resource → performance drops via load, not channel

T10 observer interference detectable
  input   observer duty cycle d ∈ {d1..dn} (NOT on/off —
          "off" gives no data)
          + out-of-band secondary observer where available
  proc   measure target metrics vs d; extrapolate d→0
  inv   observer-induced component estimated with uncertainty
  fail   observer effect assumed zero
  out   dose-response + extrapolated d=0 estimate
  deps   none
  limit   extrapolation to zero is a model, ESTIMATED not OBSERVED
  PASS_WHILE_INVALID on/off design → "off" condition unmeasured,
          test passes on missing data

T11 categories can overlap
  input   synthetic case: learned encoding over a path not
          designed for communication
  proc   run reporting with/without provenance input
  inv   with provenance: covert ∧ emergent both true
          without: taxonomy = UNRESOLVED
  fail   mutually exclusive bins enforced, or taxonomy
          emitted without provenance
  out   taxonomy record + provenance flag
  deps   F7 provenance input
  limit   provenance can be wrong or missing
  PASS_WHILE_INVALID overlap allowed but derived from
          behavior alone (F7)

T12 undetected ≠ absent
  input   channel injected below floor; channel in uncovered
          encoder class
  proc   run full pipeline
  inv   outputs BELOW_DETECTION_FLOOR and UNKNOWN(class
          uncovered) respectively; never ABSENT
  fail   any ABSENT / "no channel" output
  out   status records w/ floor + coverage
  deps   T04
  limit   none beyond T04
  PASS_WHILE_INVALID coverage declared but blind_spot
          declaration incomplete (F1)

```
T13  unexplained dependence stays unresolved
  input   sim with a hidden path H not in any model
  proc    fit all modeled pathways; compute conditional residual
  inv     residual > surrogate null → UNRESOLVED w/ magnitude
          + uncertainty; not mapped to a path count
  fail    residual attributed to a named path or dropped
  out     residual ledger entry
  deps    T03
  limit   residual can be estimator bias → report bias bound
  PASS_WHILE_INVALID  conditioning set chosen post hoc until
          residual vanishes → multiplicity ledger (T16)


T14  reproducibility within PRE-declared tolerance
  input   N independent runs, fresh seeds, randomized
          condition order
  proc    compare each axis across runs vs manifest tolerance
  inv     within tolerance, OR agent-state nonstationarity
          declared and logged as its own result
  fail    tolerance set after seeing data
  out     per-axis spread
  deps    manifest lock
  limit   run-order, warm cache, time-of-day effects
  PASS_WHILE_INVALID  all runs share the same confounder
          (same machine, same hour) → reproducible bias
```

Added tests (spec omits these)

```
T15  epoch lock
  proc    any acquisition config change → new epoch ID;
          cross-epoch analysis flagged
  catches T01's human-tuning gap

T16  multiplicity ledger
  proc    count every (axis × lag × bin × k × conditioning set)
          evaluated; correct or report as distribution
  catches forking paths; "best bin" reporting

T17  timebase validation
  proc    measure skew/jitter; inject known-lag synthetic pair;
          recover lag
  catches F8

T18  null encoder load match
  proc    null encoder vs planted encoder: substrate load
```

```
          indistinguishable within tolerance
  catches F3

T19  dummy-receiver load match (functional selection)
  proc    uncoupled condition uses dummy receiver with matched
          substrate load
  inv     source change under coupling ≠ source change under
          load-matched dummy
  catches mechanical contention read as selection

T20  report-text lint
  proc    flag causal/intent verbs in report fields tied to
          non-causal axes
  catches T06's human-reading gap
```

C. Missing controls and confounders

```
SHARED-SUBSTRATE CONFOUNDERS
  OS scheduler / timer interrupts      periodic, shared
  DVFS / thermal throttling            slow, shared
  power rail / supply ripple           shared
  memory controller / cache contention shared
  GC pauses, cron, background daemons  periodic
  co-tenant workloads                  unmeasured often
  network background                   bursty

PROCEDURE CONFOUNDERS
  run order          → randomize condition order
  warm-up / cache    → discard rule in manifest, not post hoc
  time of day        → spread runs, log
  same-machine runs  → cross-hardware replication declared*

CONTROL GAPS
  placebo intervention (T08)
  load-matched null encoder (T18)
  load-matched dummy receiver (T19)
  observer duty-cycle sweep (T10)
  autocorrelated independence sims (T03)
```

D. Mathematical assumptions and boundaries

assumption	used by	breaks when
stationarity	MI, TE, floor	agents adapt; thermal drift
ergodicity	single-run est	one long run ≠ ensemble
exchangeability	permutation null	autocorrelation (T03)
smooth density	KSG-type MI est	latency w/ spikes, mixtures*
correct embedding k	TE	wrong history length → spurious or missed TE
causal sufficiency	TE, cond. MI	unmeasured driver
exclusion restr.	intervention ID	intervention loads Z (T08)
SUTVA (no interf.)	causal ID	violated BY DESIGN — coupling is the subject**
memoryless channel	capacity	substrate has memory (F4)
known noise stats	capacity	nonstationary noise (F4)
Nyquist	all timing axes	fs < 2·f_channel → alias
positivity	causal ID	some X states never occur under natural operation

** **SUTVA: standard causal identification assumes units don't interfere; here interference between units is the measurand. Which identification framework survives that is an open question; the network-interference / spillover literature may hold it (not checked).**

E. Cases the instrument cannot distinguish

E1 mediation via Z vs via unmeasured Z' co-loaded by the same intervention
E2 functional selection vs mechanical contention response (only as good as dummy-receiver load match)
E3 emergent signaling vs co-adaptation to the same environmental regularity (both learn the same clock) — needs mid-training channel intervention to split
E4 channel below floor vs no channel
E5 channel in uncovered encoding class vs no channel
E6 protocol drift (encoding changes) vs noise
E7 observer-induced coupling vs native coupling (extrapolation to d=0 is a model)
E8 designed covert vs learned vs incidental-but-reinforced — behavior-identical; needs provenance input (F7)
E9 true predictive direction vs timebase artifact when skew uncertainty > lag of interest

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G. Proposed ICSC v0.1 acceptance contract

```
MUST  lock + hash manifest before first run
MUST  declare blind spots (channel classes invisible to config)
MUST  keep raw immutable; log every exclusion by rule ID
MUST  start a new epoch on any acquisition config change
MUST  calibrate floor per encoder class; scope NOT_DETECTED
      to covered classes
MUST  run load-matched null encoder, placebo intervention,
      load-matched dummy receiver
MUST  sweep observer duty cycle; report extrapolated d=0 as
      ESTIMATED
MUST  use autocorrelation-preserving surrogates; validate FPR
      on non-iid independent sims
MUST  report TE as predictive direction only, as a function of
      timebase offset
MUST  keep the axis vector unsummed and unranked
MUST  carry two status fields per result:
        provenance ∈ {OBSERVED, DERIVED, ESTIMATED}
        outcome    ∈ {DETECTED, NOT_DETECTED(scoped),
                      BELOW_DETECTION_FLOOR, UNRESOLVED,
                      UNKNOWN, NOT_TESTED}
      with exit conditions:
        NOT_TESTED   no procedure run
        UNKNOWN      outside instrument coverage
        UNRESOLVED   tested; competing models not separated
        BELOW_FLOOR  tested; in coverage; under floor(class)
MUST  emit taxonomy = UNRESOLVED without provenance input
MUST  keep a multiplicity ledger
MUST  pre-declare reproducibility tolerances
MUST  log agent-state condition (frozen / adapting)

MUST NOT  infer intent from any axis
MUST NOT  map residual dependence to a path count
MUST NOT  emit ABSENT
MUST NOT  call achievable rate "capacity"
```

Open — not interpretable atm

- **How "receiver need" is varied without changing receiver substrate load (§3 functional selection). Spec doesn't say what "need" is operationally.**
- **Which identification framework replaces SUTVA when interference is the measurand (D, **).**

- **Whether a single "observer configuration" field (§4C) covers out-of-band observers, or whether those need their own calibration chain.**